DRIVE: Modeling Skills at the Reasoning and Interaction Levels for Web Agents under Continual Learning

Xirui Liu, Sihang Zhou, Yanning Hou, Rong Zhou, Haoyuan Chen, Maolin He, Siwei Wang, Hao Chen, Jian Huang

Paper version: arXiv 2605.23939v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *This entanglement limits capability accumulation*

Reasoning knowledge transfers across sites; interaction knowledge is page-bound. One uniform memory loses one or the other.

Why uniform stores stall

1. **Mixed stores.** Past work blends both levels.
2. **Dilemma.** NL doesn't ground; code doesn't transfer.
3. **No routing signal.** Nothing says which level failed.

Evaluation setup

WebArena · 812 tasks | 5 domains | GPT-4.1 backbone | 20-step budget | 30/70 same-site split | Qwen2.5-7B alt

- **Online only:** functional-correctness SR.
- **Within-website 30/70 split:** same five sites throughout.

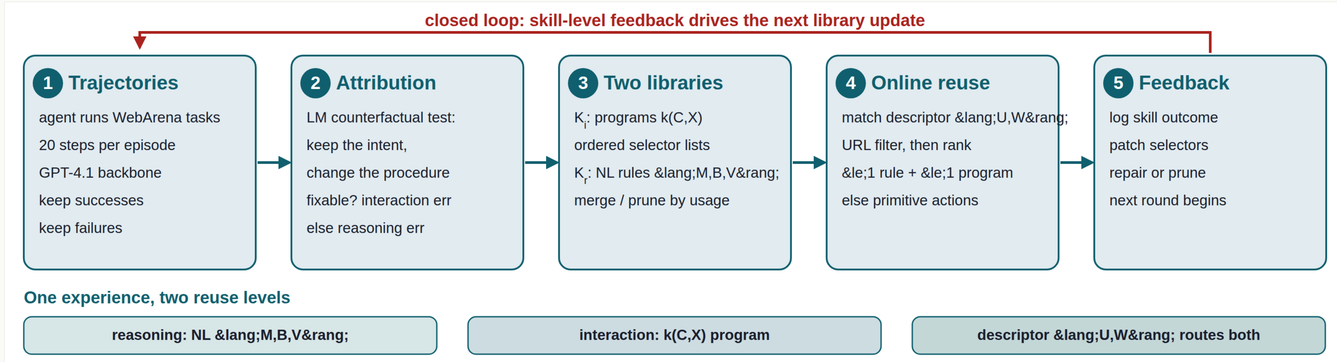
Our read (CUA Radar)

61	★★	online
score of 100	band	evidence level

- **Reports:** online eval; matched baselines; 20-step budget; cost; ablation; attribution vs humans (69.1%).
- **Flags:** skill data overlaps eval sites (AP5); no main-table variance; no code.
- Pattern C19. Authors' numbers, not rerun.

Method: two skill libraries, one closed loop

- **Sees / acts:** DOM observation + URL in; primitive actions out, or one invoked skill (desktop web).
- **Two levels:** NL reasoning rules ⟨M,B,V⟩ say *what*; programmatic interaction skills k(C,X) execute *how*.
- **Signal:** task outcome + skill feedback; counterfactual attribution routes each failure to the right library.



$$k^i(C, X) \rightarrow (\alpha, e)$$

interaction skill: page context + args in; action trace + outcome out

Key results: WebArena success rate (%)

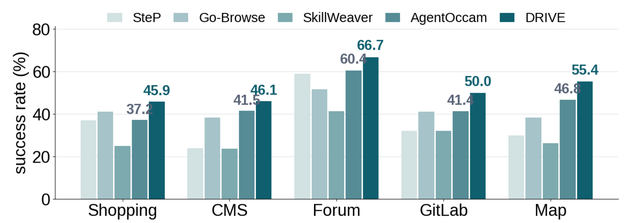


Table 1, redrawn: all methods GPT-4.1, online SR.

52.8% average vs 45.5% AgentOccam (+7.3 pp, +16.0% relative)

Ablation: WebArena average (%)

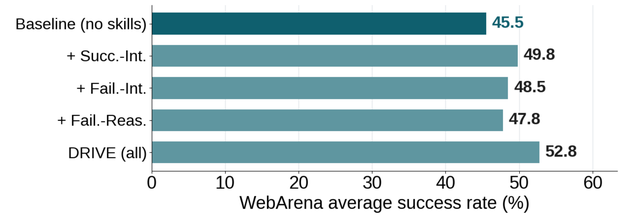


Table 2, redrawn: which induction pathway is added to the skill-free baseline.

SO WHAT → *Successes alone +4.3 pp; each failure-routed pathway adds less alone; the full stack +7.3. Table 4: the two levels cut different error types.*

Headline Numbers

52.8%
WebArena success rate
vs 45.5% AgentOccam

- **+7.3** pp vs AgentOccam
- **69.1%** attribution accuracy
- **12.5%** Qwen2.5-7B avg SR

The paper's own controls

CONTROL	RESULT	CONTEXT
Same-site split	30% train / 70% test	within-website (AP5)
Attribution quality	54.0 human vs 46.0 none	150-task subset
Open backbone	8.2 → 12.5 avg	Qwen2.5-7B, +52% rel.
Cost per success	164.7k → 151.9k	−7.8% input/succ.

Reversed routing: 44.7% vs 46.0% no-skill

Takeaway

Dual-level skills + failure attribution: 45.5% → 52.8% WebArena (GPT-4.1), on the same sites.

42 → 251 train traj. | same-site split | no code found

Scan to read the full paper

arxiv.org/abs/2605.23939v1